

# Change point methods benchmarking methodology

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## 1 Motivation

Consider a multi-variate time series  $X_1, \dots, X_N$  that is assumed to follow one distribution before the unknown moment of time  $\tau$  and another distribution afterwards:

$$\begin{aligned} X_i &\sim \mathbb{P}, & i \leq \tau, \\ X_i &\sim \mathbb{Q}, & i > \tau. \end{aligned}$$

At every time  $t$ , we form two adjacent windows of equal length  $w$ :

- **reference window:**  $A_t = \{X_{t-2w}, \dots, X_{t-w-1}\}$
- **test window:**  $B_t = \{X_{t-w}, \dots, X_{t-1}\}$

We then test the hypothesis

$$H_0 : P_A = P_B \text{ vs } H_1 : P_A \neq P_B$$

We compute a discrepancy statistic

$$D_t = D(A_t, B_t)$$

If  $D(A_t, B_t) > \gamma$ , where  $\gamma$  is a chosen threshold, we reject  $H_0$  and return a changepoint.

## 2 General sliding-window framework

In our simulations, we compare three non-parametric methods of change point detection. Each of them corresponds to a two-samples statistical test as described in the first section.

1. TEWMA based on running-window Euler Characteristic Curve explained before
2. Kernel MMD with a fixed kernel width  $\sigma$ .  
Let  $k(X, Y)$  be a spsd kernel. The distribution  $\mathbb{P}$  is presented as

$$\mu_P = \mathbb{E}[k(X, \cdot)].$$

Similarly,

$$\mu_Q = \mathbb{E}[k(Y, \cdot)].$$

We then define MMD distance as

$$\text{MMD}(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}}.$$

in the RKHS associated to the gaussian kernel. By injectivity of the gaussian kernel, we have that

$$\text{MMD}(P, Q) \iff P = Q$$

In our algorithm, we use the empirical version of MMD. Namely, given two samples,

$$X = \{x_i\}_{i=1}^m, \quad Y = \{y_j\}_{j=1}^n.$$

, we define the empirical MMD as

$$\widehat{\text{MMD}}^2(P, Q) = \frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m k(x_i, x_j) + \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n k(y_i, y_j) - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(x_i, y_j).$$

Thus, in the MMD algorithm, we have

$$D(A_t, B_t) = \widehat{\text{MMD}}^2(A_t, B_t)$$

with  $A_t$  and  $B_t$  defined as before, and  $k(\cdot, \cdot)$  is the Gaussian kernel

$$k(x, y) = \exp(-\gamma \|x - y\|^2)$$

,  $\gamma$  being a hyperparameter.

3. Energy-distance approach based on pairwise distances between two samples. This method uses another two-sample statistic, called **Energy distance**. Namely, Energy Distance compares distributions directly through Euclidean distances between observations. For two distributions  $P$  and  $Q$ ,

$$\mathcal{E}(P, Q) = 2 \mathbb{E} \|X - Y\| - \mathbb{E} \|X - X'\| - \mathbb{E} \|Y - Y'\|.$$

and then

$$D(P, Q) = \mathcal{E}(X, Y)$$

### 3 Numeric simulations

We provide the code for all three approaches in three corresponding files in the folder `changepoint`. To facilitate the numeric computations, each function accepts a sorted list of thresholds and then returns the corresponding detection delay for each of them; namely, for each threshold  $\gamma$ , it returns

$$\inf_{t \geq T} \{D(A_t, B_t) > \gamma\}$$

for the specified algorithm. Here, we prefer to take a reference point  $T$  to be able to detect the Average Run Length (ARL) and the Detection Delay (DD), which are defined correspondingly as  $\text{ARL} = \inf\{t : D_t > \gamma\}$  and  $\text{DD} = \inf\{t > \tau : D_t > \gamma\}$  where  $\tau$  is the changepoint. Thus, we can compute both ARL and DD with one function by setting  $T = -1$  and  $T = \tau$  correspondingly.

### 4 Monte-Carlo trials

To obtain meaningful information about the nature of our algorithms, we run  $M = 100$  Monte-Carlo trials for each pair of null and alternative distribution. Namely, we average ARL and DD for each threshold (the set of thresholds remains the same throughout epochs), dividing the sum of each iteration by the number of epochs.